TECHNICAL EVALUATION — SUBMERSIBLE PUMP

1. Evaluation output

Field	Value
Report	submersible_pump_EN201222034S.pdf (89 pages)
Standards	IEC 60335-1:2010 (Part 1); IEC 60335-2-41:2021 (Part 2 2-41)
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2. Report identity

Field	Value
Report number	EN201222034S
Laboratory	EMTEK(NINGBO) CO., LTD.
Report date	Dec 28, 2025
Product declared	Submersible pump
Declared model	SPC55068
Standards claimed	IEC 60335-1:2010+A1:2013+A2:2016; IEC 60335-2-41:2025 on page 2; IEC 60335-2-41:2012 on page 3
Standards evaluated	US IEC 60335-1:2010; IEC 60335-2-41:2012 as reproduced in ILNAS-EN IEC 60335-2-41:2021

3. Summary

Field	Value
Final verdict	REPORT CORRECTION / REPLACEMENT REQUIRED — PRODUCT COMPLIANCE NOT DETERMINED — CERTIFICATION HOLD
Reason	The tested model and electrical test basis conflict with the declared product, and several applicable safety assessments lack substantiated evidence. The claimed 2025 particular part and Part 1 amendments were not supplied, so those claims cannot be evaluated.
Recommended action	not stated
Conclusion	The report does not demonstrate compliance of SPC55068 with the supplied general and particular requirements. A replacement report must establish sample identity and ratings, resolve the conflicting test conditions, and provide the missing marking, immersion, cord, earthing, insulation and heat-and-fire evidence; no product non-conformity is established solely from the mismatched measurements.
Findings	22 total — Critical 4, High 16, Medium 2, Low 0

4. Findings

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
Critical	5.2; 7.1	The test evidence cannot be reliably attributed to the declared model.	The declared model is SPC55068, but the summary states that full tests were performed on model 4TSC4.0-96-110/1000. No relationship between the models is established.	Provide a replacement report identifying the tested sample and establishing its relationship to SPC55068.	PDF pp.2–3	Report: verified p.3 Standard: verified in IEC 60335-1:2010 p.26
Critical	7.1	The voltage used to identify the appliance and assess its marking is inconsistent with the declared rating.	The declared rating is 220–240 V, but the marking assessment records 96 V.	Verify the actual marking and correct the report to identify the applicable rated voltage.	PDF pp.2–3,6	Report: no quote given Standard: verified in IEC 60335-1:2010 p.29
Critical	10.1	The reported measurement is not evaluated against the declared model's rating and rated voltage.	The declared rated input is 550 W, while the power-input table uses 1,000 W and records 995.2 W at 72 V.	Test and report the power input of SPC55068 against its verified rating.	PDF pp.2–3,79	Report: no quote given Standard: verified in IEC 60335-1:2010 p.38
Critical	5.8.1; 5.8.2; 11.5; 13.1; 16.2	The recorded supply and test voltage do not establish testing at the applicable conditions for the declared rating.	The report declares 220–240 V, 50 Hz, but identifies the supply as DC and records 76.3 V for heating and leakage-current tests.	Verify the supply type and repeat or substantiate the affected tests at the applicable voltage and frequency.	PDF pp.2–4,14–15,79–80	Report: no quote given Standard: verified in IEC 60335-1:2010 p.27
High	5.101; 25.14	Pumps are tested as portable unless fixed; the particular part subjects portable pumps, except specified types, to the flexing test.	Portable and fixed classifications are not identified in the particulars; testing as a portable pump and the cord-flexing test are marked N/A.	Establish the pump classification and perform the applicable cord-flexing test.	PDF pp.4,6,43	Report: verified p.6 Standard: verified in IEC 60335-2-41:2021 p.13
High	7.1	A submersible pump above 50 W must be marked with its maximum operating depth in metres.	Maximum operating depth is recorded as N/A for the declared 550 W submersible pump, and no depth is shown on the reproduced marking plate.	Verify and document the maximum operating depth marking on the tested pump.	PDF pp.2–4,7	Report: verified p.4 Standard: verified in IEC 60335-2-41:2021 p.14
High	7.1; 5.7; 11.7	The pump must be marked with a maximum liquid temperature; liquid temperature determines the specified test conditions.	The summary gives maximum liquid temperature as "<45□", but the reproduced marking plate gives no maximum liquid temperature and no actual liquid test temperature is recorded.	Verify the temperature marking and document the actual liquid temperatures used in testing.	PDF pp.3,6–7,14,79	Report: verified p.3 Standard: verified in IEC 60335-2-41:2021 p.14
High	7.12.1	Installation instructions must state maximum total head for pumps rated above 50 W.	Maximum total head for the 550 W pump is passed by reference to marking details, but no value is shown on the reproduced plate or elsewhere in the supplied record.	Provide the instructions and record the maximum total head in metres.	PDF pp.3,10	Report: verified p.10 Standard: verified in IEC 60335-2-41:2021 p.14
High	15.1.2	The supplied particular part specifies the pressure based on maximum depth or 15 m, if higher, for the 1.3-times branch.	The report records a 24 h immersion pass and a 1.3-times-pressure branch, but gives no maximum operating depth or actual test pressure; its alternative reference is 5 m rather than 15 m.	Establish the marked depth and document a correctly selected immersion pressure and test result.	PDF pp.4,17	Report: verified p.17 Standard: verified in IEC 60335-2-41:2021 p.16
High	19.7	The locked-rotor test is performed at rated voltage for the applicable period, with winding temperatures assessed accordingly.	The locked-rotor test records 116.6 V DC, inconsistent with the declared 220–240 V, 50 Hz rating; test duration is not recorded.	Repeat or substantiate the locked-rotor test at the verified rated supply and document its duration.	PDF pp.2,20–21,81	Report: verified p.81 Standard: verified in IEC 60335-1:2010 p.54
High	19.7; 24.5	Motors with an auxiliary-winding capacitor require the specified locked-rotor capacitor fault tests, subject to the stated P2 exception.	Capacitor checks are marked P under 24.5, but the locked-rotor capacitor open- and short-circuit checks are marked N/A without explanation.	Establish whether an auxiliary-winding capacitor is fitted and document the applicable fault tests or exemption.	PDF pp.20,40	Report: verified p.20 Standard: verified in IEC 60335-1:2010 p.54

Priority	Clause	Normative requirement	Finding reason	Required action	Pages	Evidence
High	24.1.4	Applicable controls and motor protectors require evaluation under the stated component or Annex D provisions.	The particulars indicate a self-resetting thermal cut-out, while automatic-control and thermal-motor-protector assessments are marked N/A; the protector's role in compliance is not established.	Identify the cut-out and document its safety function and applicable evaluation.	PDF pp.4,38,62	Report: verified p.4 Standard: verified in IEC 60335-1:2010 p.77
High	25.1; 25.3	The particular requirements apply to submersible pumps other than class III.	The pump is declared class I and submersible, but the particular-part entries for a flexible cord and cord fitted with a plug are marked N/A; a general plug entry is passed.	Confirm and document the fitted flexible cord and plug for the tested pump.	PDF pp.4,40–41	Report: verified p.40 Standard: verified in IEC 60335-2-41:2021 p.19
High	25.7; 25.8	The supplied standards specify cord construction, conductor area and minimum length for applicable outdoor submersible pumps.	Outdoor use is indicated in the particulars, but cord type, conductor cross-sectional area and outdoor cord length are not established; the relevant entries are N/A.	Identify and verify the cord type, conductor area and length against the applicable requirements.	PDF pp.4,41–42	Report: verified p.42 Standard: verified in IEC 60335-2-41:2021 p.19
High	25.10; 27.3	Class I supply cords require a green/yellow earthing core, and the conductor arrangement must preserve earthing if the cord slips.	The report identifies a class I appliance with a supply cord, but protective-conductor colour and the conductor arrangement on cord displacement are marked N/A.	Inspect and record the protective conductor and its arrangement at the cord anchorage.	PDF pp.4,43,49	Report: verified p.43 Standard: verified in IEC 60335-1:2010 p.82
High	27.5	The standard requires a specified test current and a calculated resistance not exceeding 0.1 Ω.	Earth continuity is marked P, but no measured resistance, test current or tested connection is recorded.	Provide the earth-continuity test conditions and measured resistance for each applicable accessible metal part.	PDF p.49	Report: verified p.49 Standard: verified in IEC 60335-1:2010 p.91
High	29.1; 29.2	The insulation assessment must use the applicable rated impulse and working voltages, pollution degree and material group.	The clearance table uses a 500 V impulse rating and the basic-creepage table uses a working voltage of ≤50 V, despite the declared 220–240 V rating; basic clearance and creepage are reported as >2.0 mm.	Reassess and report each applicable insulation distance using the verified electrical ratings and construction.	PDF pp.2,51–54,83–85	Report: verified p.83 Standard: verified in IEC 60335-1:2010 p.97
High	29.3; 25.13	Supplementary insulation must be assessed by one of the specified routes.	The report passes the inlet bushing as supplementary insulation and states that insulation was checked by measurement, but gives no insulation thickness or alternative assessment result.	Provide the bushing insulation assessment and its measured thickness or applicable alternative test evidence.	PDF pp.43, 54–55,82	Report: verified p.82 Standard: verified in IEC 60335-1:2010 p.101
High	30.1	Relevant non-metallic and insulating parts require the heat-resistance assessment where deterioration could impair compliance.	The entire resistance-to-heat assessment is marked N/A and the appended table contains no part-specific test results or justification.	Identify the relevant parts and provide the required heat-resistance evidence or a justified exclusion for each.	PDF pp.55–56,87	Report: verified p.55 Standard: verified in IEC 60335-1:2010 p.102
High	30.2; 30.2.1; 30.2.3	The particular part selects 30.2.1 for the specified enclosed submersible-pump construction and RCD instruction, or 30.2.3 for other pumps.	Fire-resistance clauses and the appended table are marked N/A or blank, without establishing the construction and RCD-instruction conditions for the particular-part route.	Establish the applicable fire-test route and provide the relevant part or material evidence.	PDF pp.56–59,87	Report: verified p.56 Standard: verified in IEC 60335-2-41:2021 p.20
Medium	—	The conflicting claim cannot be reconciled or evaluated against the unsupplied editions and amendments.	The test specification claims IEC 60335-2-41:2025, while the compliance summary claims IEC 60335-2-41:2012; the claimed Part 1 amendments and 2025 particular part were not supplied.	Provide the claimed standards and amendments and issue a report with a consistent test specification.	PDF pp.2–3	Report: verified p.2 Standard: requirement not in supplied standards
Medium	—	The page's association with the report is uncertain, including its continuation of the creepage table.	The footer on PDF page 86 identifies report EN200622010S rather than EN201222034S.	Replace or verify the page under the correct report number.	PDF p.86	Report: verified p.86 Standard: requirement not in supplied standards